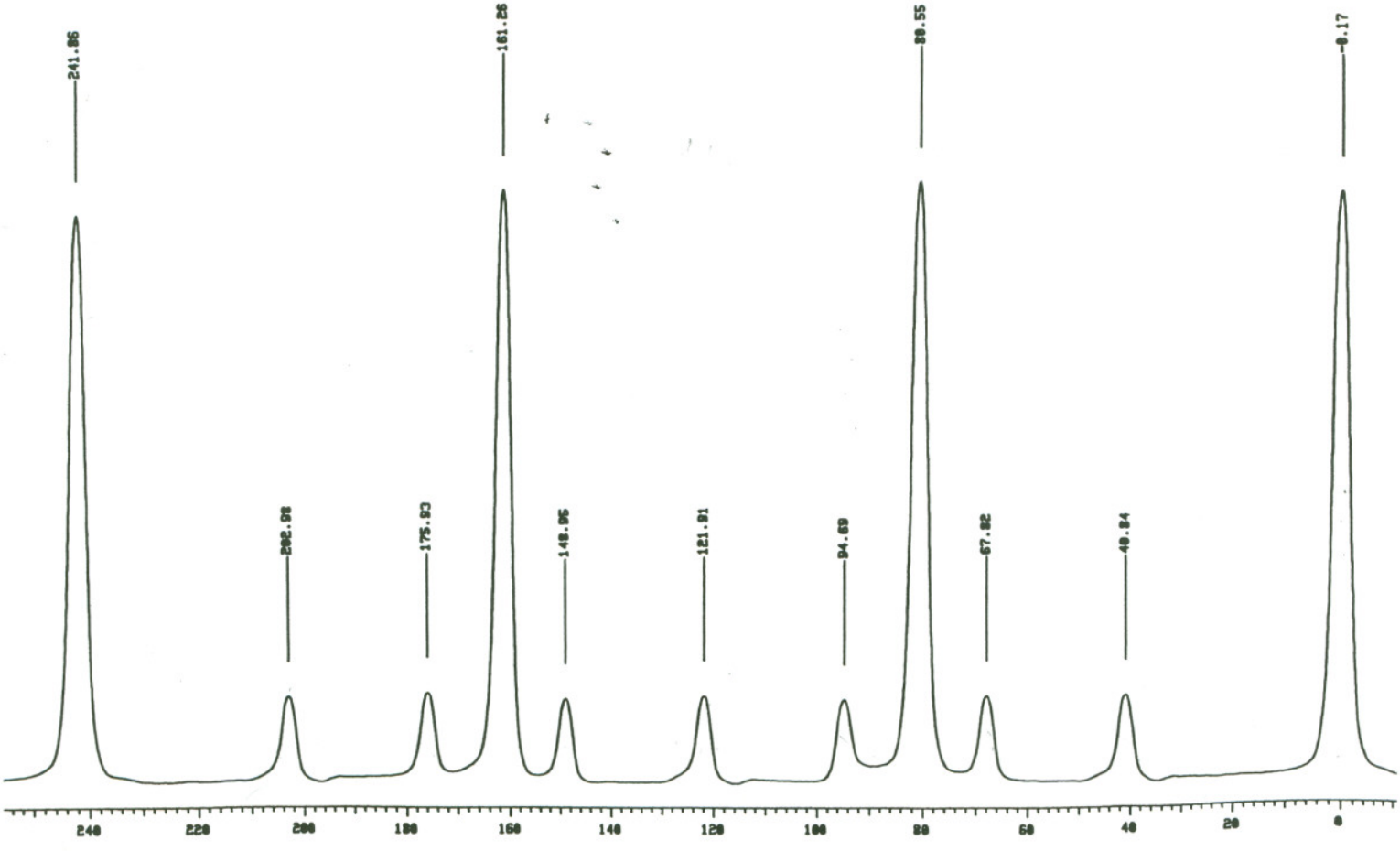
Physical Cume
March 1, 2002

Two isotopes of the same element (A and X) form the compound with hydrogen, AH_4 and XH_4. Both isotopes, A and X, are NMR active. The spin quantum number for isotope A is I = 3. Using the ^1H NMR spectrum shown below (the scale is in Hz), determine the spin quantum number of isotope X. Discuss the magnitude of the spin-spin coupling constants in the AH_4 and XH_4 molecules. Which isotope, A or X, has the highest resonance frequency at a magnetic field of 11.75 T. Which of the isotopes, A or X, is the more abundant? Draw the theoretical NMR spectrum of isotopes A and X (^1H I =1/2). Determine the ^1H chemical shift, in Hz, for the AH_4 and XH_4 molecules and explain your finding.